

GAOPING, China

WMO: 574110

Lat: 30.7456N

Lon: 106.1319E

Elev: 347

StdP: 97.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	2.7	-3.9	2.8	7.4	-1.2	3.6	7.1	5.5	5.4	4.5	6.0	1.1	20	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	36.5	24.8	35.1	24.8	33.7	24.6	27.1	32.0	26.5	31.3	26.0	30.6	2.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	22.2	29.3	25.3	21.4	28.9	24.8	20.7	28.4	88.6	32.4	85.7	31.2	83.2	30.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.0	4.0	3.3	DB	-0.4	38.3	1.4	1.6	-1.4	39.5	-2.2	40.4	-2.9	41.3	-4.0	42.5
			WB	-1.0	28.0	1.5	0.9	-2.1	28.6	-3.0	29.2	-3.8	29.7	-4.9	30.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	6.8	9.4	13.8	18.9	22.5	25.2	28.3	28.4	23.6	18.5	13.5	8.2	(6)
(7)		DBStd	7.93	2.06	2.71	3.52	3.67	2.86	2.57	2.58	3.30	3.44	2.93	2.84	2.18	(7)
(8)		HDD10.0	215	102	41	6	1	0	0	0	0	0	4	62	(8)	
(9)		HDD18.3	1289	358	252	147	36	3	0	0	1	34	145	313	(9)	
(10)		CDD10.0	3182	2	22	124	268	387	454	567	569	408	262	111	7	(10)
(11)		CDD18.3	1212	0	0	6	53	132	204	309	311	159	38	1	0	(11)
(12)		CDH23.3	11290	0	0	34	337	868	1634	3537	3621	1129	129	1	0	(12)
(13)		CDH26.7	4854	0	0	4	92	268	571	1631	1826	435	27	0	0	(13)
(14)	Wind	WSAvg	1.3	1.1	1.2	1.4	1.5	1.5	1.3	1.4	1.4	1.2	1.1	1.0	(14)	
(15)	Precipitation	PrecAvg	1026	17	17	31	74	131	131	176	135	166	92	38	18	(15)
(16)		PrecMax	1433	46	56	62	135	245	295	474	268	350	241	83	65	(16)
(17)		PrecMin	684	1	4	5	21	55	45	70	12	34	23	9	2	(17)
(18)		PrecStd	169	10	12	16	28	54	61	97	73	83	50	19	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	19.4	26.9	32.0	33.6	34.8	37.5	38.6	37.0	29.8	22.2	14.9	(19)
(20)		MCWB	9.3	12.6	17.7	20.1	21.2	23.6	25.4	24.7	24.0	20.5	16.3	10.6	(20)	
(21)		2%	DB	12.4	16.7	23.8	29.3	31.5	33.1	35.9	37.0	33.7	26.6	20.4	13.3	(21)
(22)		MCWB	8.5	11.3	15.5	19.0	20.7	23.5	25.4	24.7	23.6	19.8	15.5	10.4	(22)	
(23)		5%	DB	11.0	15.0	21.5	27.0	29.6	31.5	34.7	35.7	31.0	24.3	18.8	12.2	(23)
(24)		MCWB	8.1	10.8	14.7	18.3	20.7	23.5	25.5	24.7	23.0	19.2	15.2	9.6	(24)	
(25)		10%	DB	9.8	13.5	19.5	25.0	27.9	30.0	33.3	34.1	29.0	22.5	17.4	11.2	(25)
(26)		MCWB	7.4	10.1	13.9	17.9	20.3	23.4	25.3	24.7	22.6	18.7	14.8	9.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	13.4	19.1	22.2	23.9	26.9	28.1	27.6	26.4	22.2	17.7	12.6	(27)
(28)		MCDWB	12.5	16.9	24.2	28.5	28.6	30.2	33.0	33.3	32.0	25.9	20.2	13.6	(28)	
(29)		2%	WB	9.3	12.3	16.7	20.7	22.9	25.7	27.3	26.8	25.4	21.0	16.7	11.2	(29)
(30)		MCDWB	11.3	15.1	21.3	25.8	27.8	29.6	32.2	32.0	29.8	24.4	18.7	12.4	(30)	
(31)		5%	WB	8.5	11.4	15.5	19.7	22.1	25.0	26.8	26.2	24.4	20.1	16.0	10.2	(31)
(32)		MCDWB	10.2	14.1	20.0	24.2	26.6	28.8	31.5	31.3	28.6	22.9	17.8	11.4	(32)	
(33)		10%	WB	7.8	10.5	14.4	18.7	21.4	24.3	26.3	25.7	23.5	19.3	15.2	9.5	(33)
(34)		MCDWB	9.5	13.0	18.5	23.4	25.6	28.0	30.7	30.7	27.7	21.8	16.9	10.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	4.4	5.3	6.6	7.4	7.4	6.5	7.0	7.6	6.0	4.9	4.5	3.9	(35)	
(36)		5% DB	MCDBR	7.2	8.3	10.6	11.2	10.9	9.6	9.2	10.1	9.7	8.6	7.3	5.8	(36)
(37)		MCWBR	3.8	3.8	4.0	3.5	3.1	2.8	2.3	2.3	2.6	3.0	3.4	3.2	(37)	
(38)		5% WB	MCDBR	5.7	6.8	9.0	9.0	8.1	7.0	7.9	8.4	7.7	6.7	5.4	4.0	(38)
(39)		MCWBR	3.4	3.5	3.9	3.6	3.2	2.8	2.7	2.7	2.7	2.8	3.0	2.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.763	0.807	0.812	0.781	0.722	0.692	0.678	0.710	0.734	0.813	0.828	0.797	(40)	
(41)		taud	1.361	1.313	1.338	1.422	1.559	1.638	1.682	1.605	1.538	1.382	1.348	1.362	(41)	
(42)		Ebn at Noon	468	503	550	599	642	659	667	637	594	493	426	419	(42)	
(43)		Edh at Noon	284	323	335	318	278	256	244	261	270	297	285	272	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.64	2.04	3.02	3.92	4.11	4.03	4.88	4.52	3.17	2.25	1.75	1.46	(44)	
(45)		RadStd	0.23	0.34	0.35	0.51	0.54	0.56	0.70	0.69	0.48	0.38	0.23	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	-0.74	N/A	-0.40	N/A	-24	N/A	N/A	N/A
		N/A	N/A	-0.91	N/A	-0.24	-0.29	N/A	N/A	+76	N/A

Nomenclature: See separate page